



Statistical inference of a gene regulatory network in Arabidopsis under the combination of climate change and nutritional starvations

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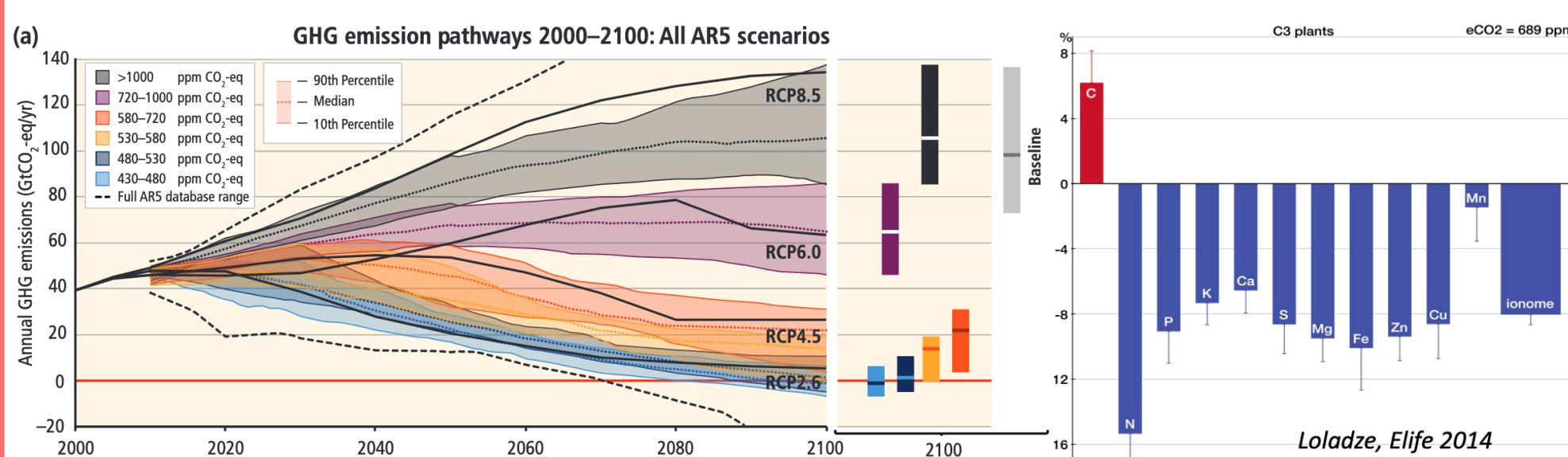
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01. Elevation of CO2 levels : a major threat to human health

ANTHROPIC CO2 EMISSIONS AS A CHALLENGE TO AGROECOLOGICAL

As plants metabolise CO₂ through **photosynthesis**, an elevation in atmospheric CO₂ could be seen as beneficial, leading to an increase of plant biomass worldwide. However, two repercussions of elevated CO₂ appear to be deleterious :

- **CO₂ acclimation causes** biomass increase to be lower than theoretically predicted^[1]
- Elevated CO₂ leads to a marked **depletion in the content of most nutrients in crops**, deteriorating their nutritional value^[2]. This exposes populations relying on primary agriculture to deficiencies in the future.^[3]



Intergovernmental Panel on Climate Change predictions, 2014 : according to the the most likely scenario, CO₂ emissions could be doubled in 2100

Relative change in C3 plants ionomes under elevated CO₂

UNDERSTANDING REGULATORY MECHANISMS INVOLVED IN MINERAL STATUS DEPLETION

In order to preserve food security for in a context of **sustainable agriculture with low inputs**, it is crucial to understand how elevated CO₂ impacts regulatory circuits associated with plant mineral nutrition.

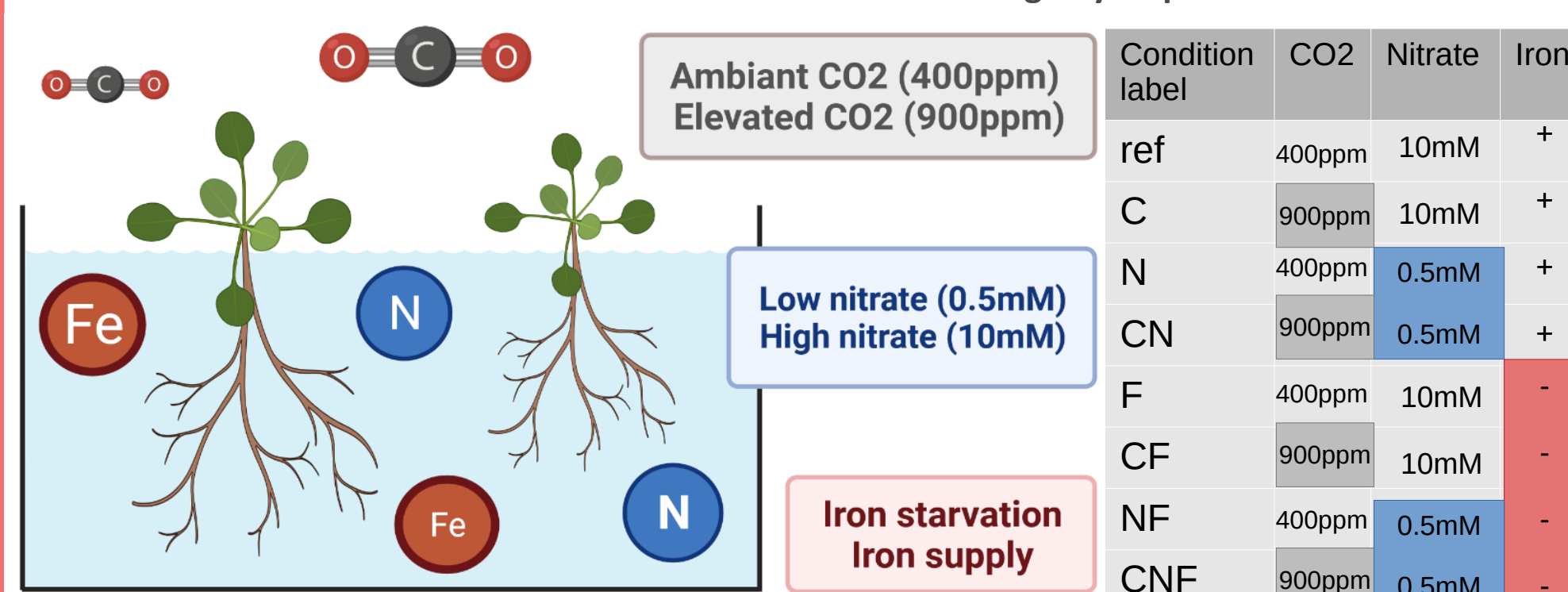
Although studies on physiological mechanisms have shown that elevated CO₂ and mineral nutrition maintain a complex interplay^[1,4,5,8], the molecular processes driving them remain unclear.

Studies on plant populations identified **genetic variation** in the response of the mineral status to elevated CO₂^[6], but **no key regulator genes have been discovered yet**. We present here an experimental design and a panel of statistical analyses to uncover pathways orchestrating nutrient uptake in roots under high CO₂.

02. Combinatorial dataset

EXPERIMENTAL DESIGN OF A TRANSCRIPTOMIC DATASET

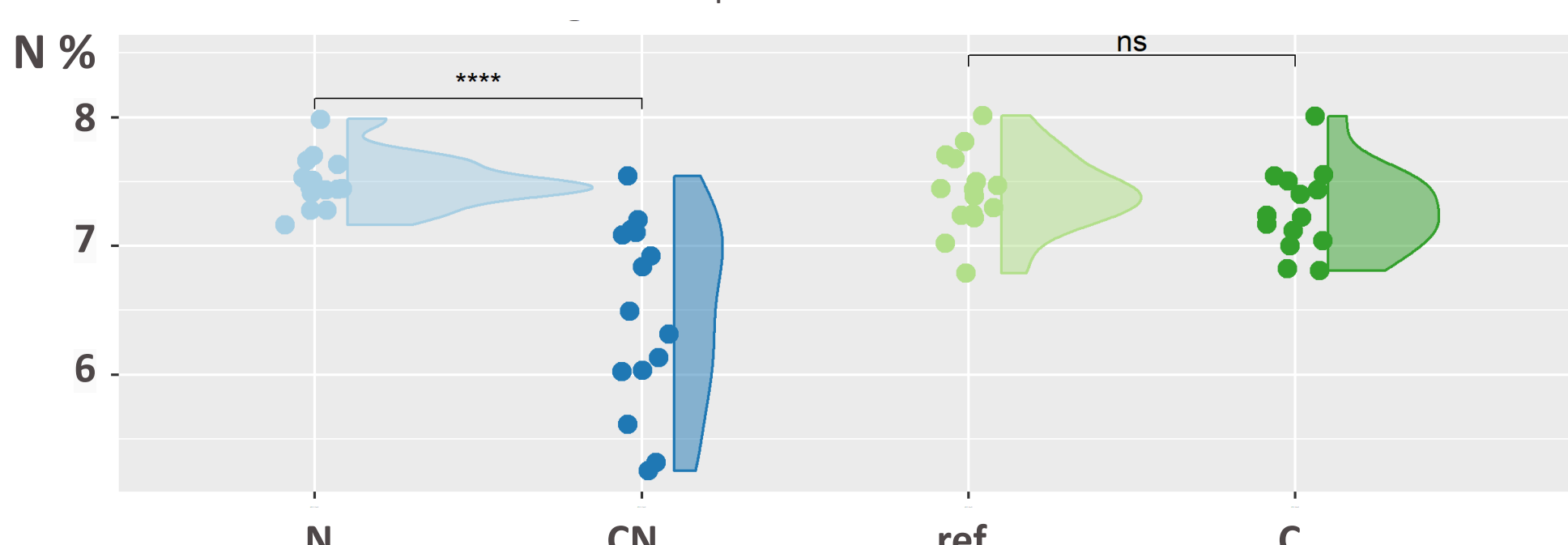
In order to investigate the impact of CO₂ on mineral nutrition in roots, we generated a transcriptomic datasets in which Arabidopsis plants were exposed to elevated CO₂, **low nitrate** conditions and **iron starvation** in a combinatorial fashion through hydroponic culture.



Schematic view of the combinatorial experiment and conditions labels

PHENOTYPIC OBSERVATION OF NITROGEN STATUS DEPLETION

With this design, we could confirm the **significant decrease in nitrogen percentage** under elevated CO₂ in foliar tissues, only in the case where **nitrate nutrition was low** (N vs CN). This indicator of the plant mineral status is one of the most clearly impacted. We could also confirm **significant increase in biomass** in elevated CO₂ as compared to ambient.



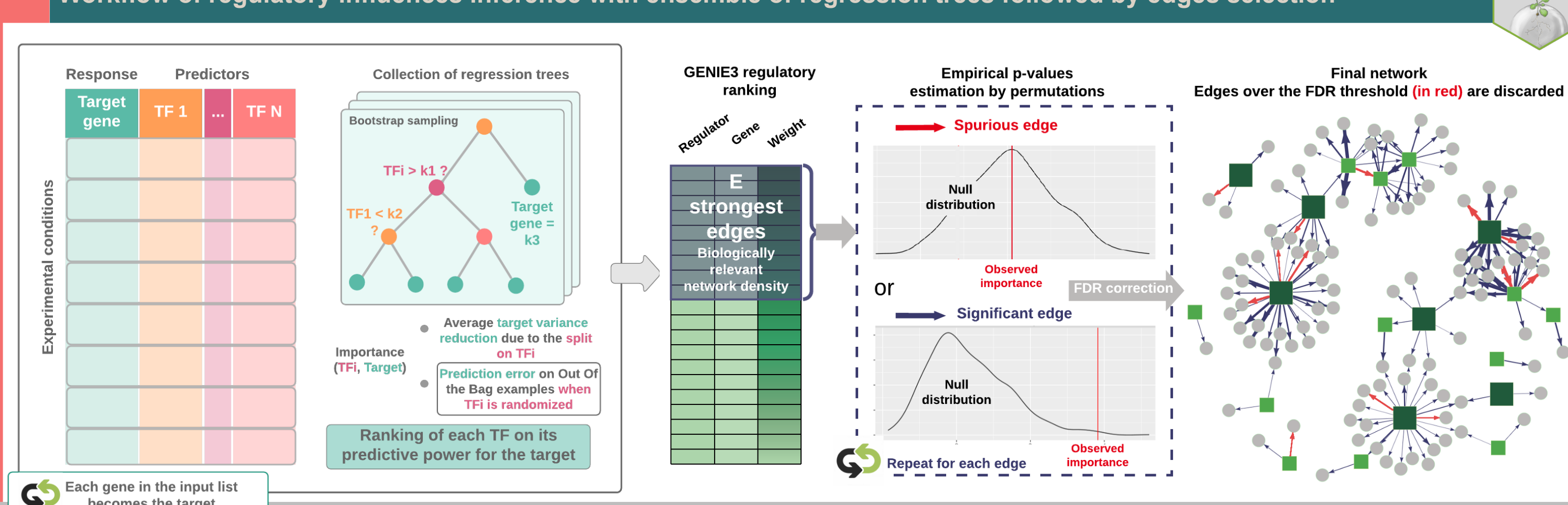
Nitrogen percentage in shoots for the ref, C, N and CN conditions.

04. Network inference method

HOW CAN WE RECONSTRUCT GENE REGULATORY INTERACTIONS FROM RNA-SEQ EXPRESSION PROFILES ?

The chosen approach, GENIE3, uses Random Forests, a machine learning method based on the inference of a collection of regression trees. This regression framework allows to infer oriented edges from regulators to targets, weighted by the predictive power of each regulator on the target gene expression. It has the advantage of being a non-parametric procedure, requiring very few modelling or biological priors, while being able to capture interactions and high order combinatorics between regulators on large scale datasets. To select statistically significant pairs in the final network, we designed a procedure of empirical tests on regulatory edges using permutations on Random Forests importance metrics.

Workflow of regulatory influences inference with ensemble of regression trees followed by edges selection



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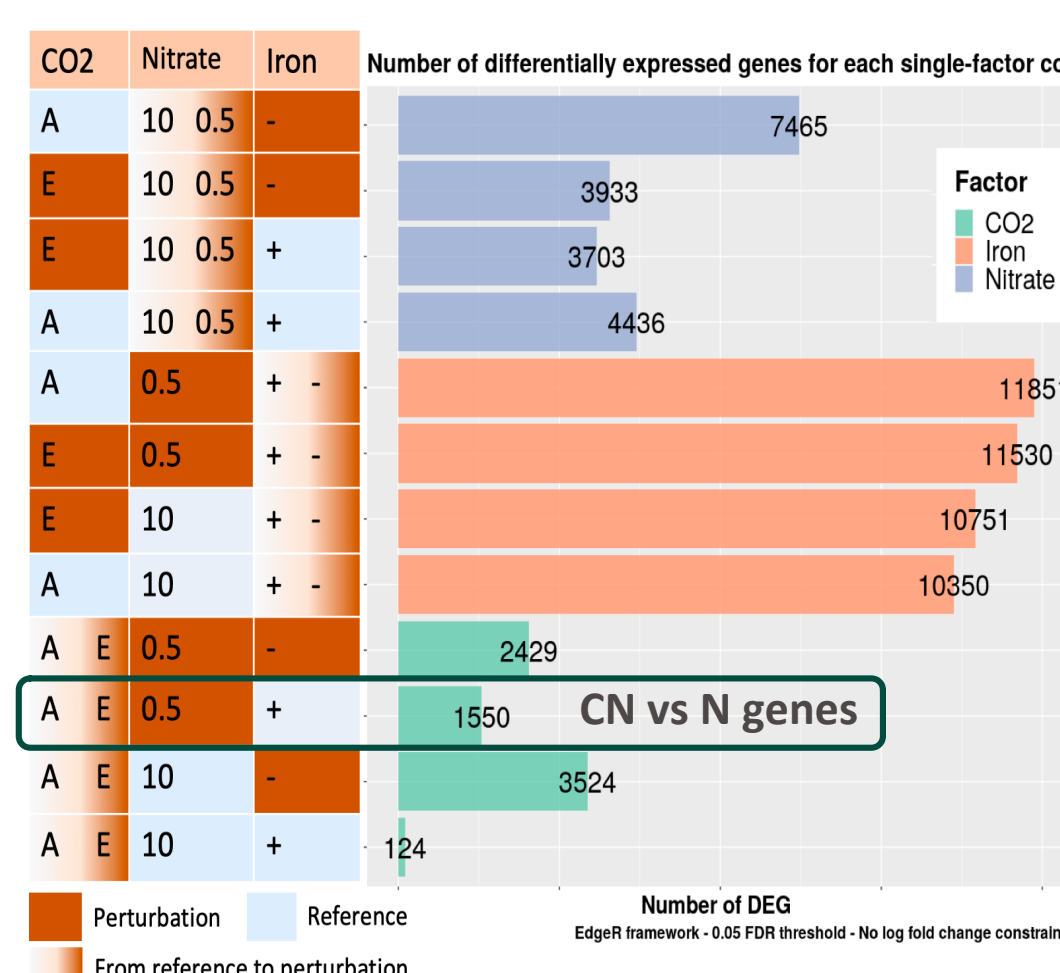
03. Elevated CO2 deregulates the response to nutritional starvations: clustering and network inference

GLOBAL TRANSCRIPTOME ANALYSIS

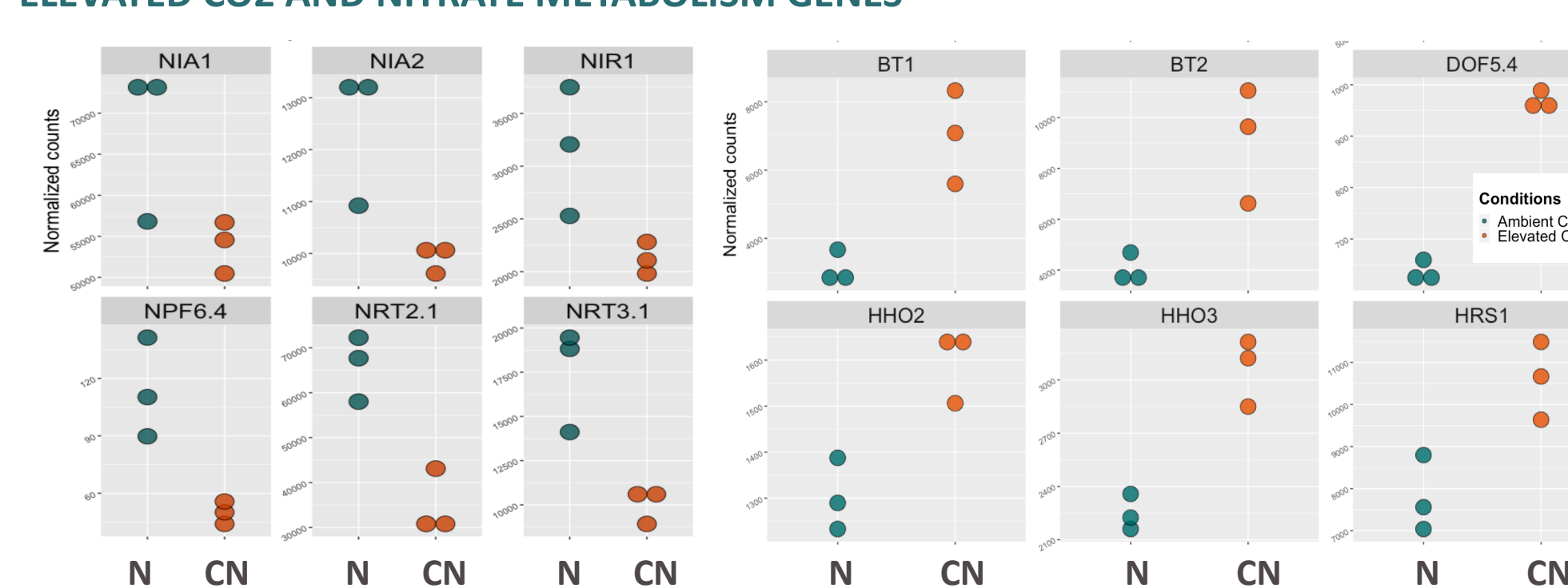
Dimensionality reduction and differential expression analyses were carried out to quantify the impact of each simple and combined perturbation on gene expression

MAIN OBSERVATIONS

- **Iron starvation** is the most influent perturbation on gene expression (high number of DEGs and PCA analysis, not shown)
- The effect of CO₂ on gene expression is **very slim when there is no nutritional starvation**
- We chose to focus on the **CN vs N genes**, that are the genes differentially expressed by elevated CO₂ when the plant are, in both transcriptomes, under low nitrate conditions.



ELEVATED CO2 AND NITRATE METABOLISM GENES

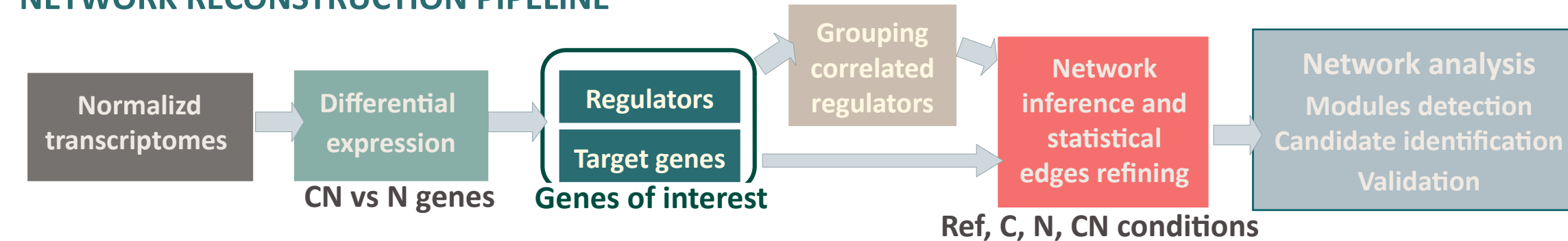


Genes involved in nitrate uptake, transport and assimilation are repressed under elevated CO₂ and low nitrate, while a number of their known repressors are upregulated

GENE REGULATORY NETWORK OF THE RESPONSE TO ELEVATED CO2 UNDER LOW NITRATE

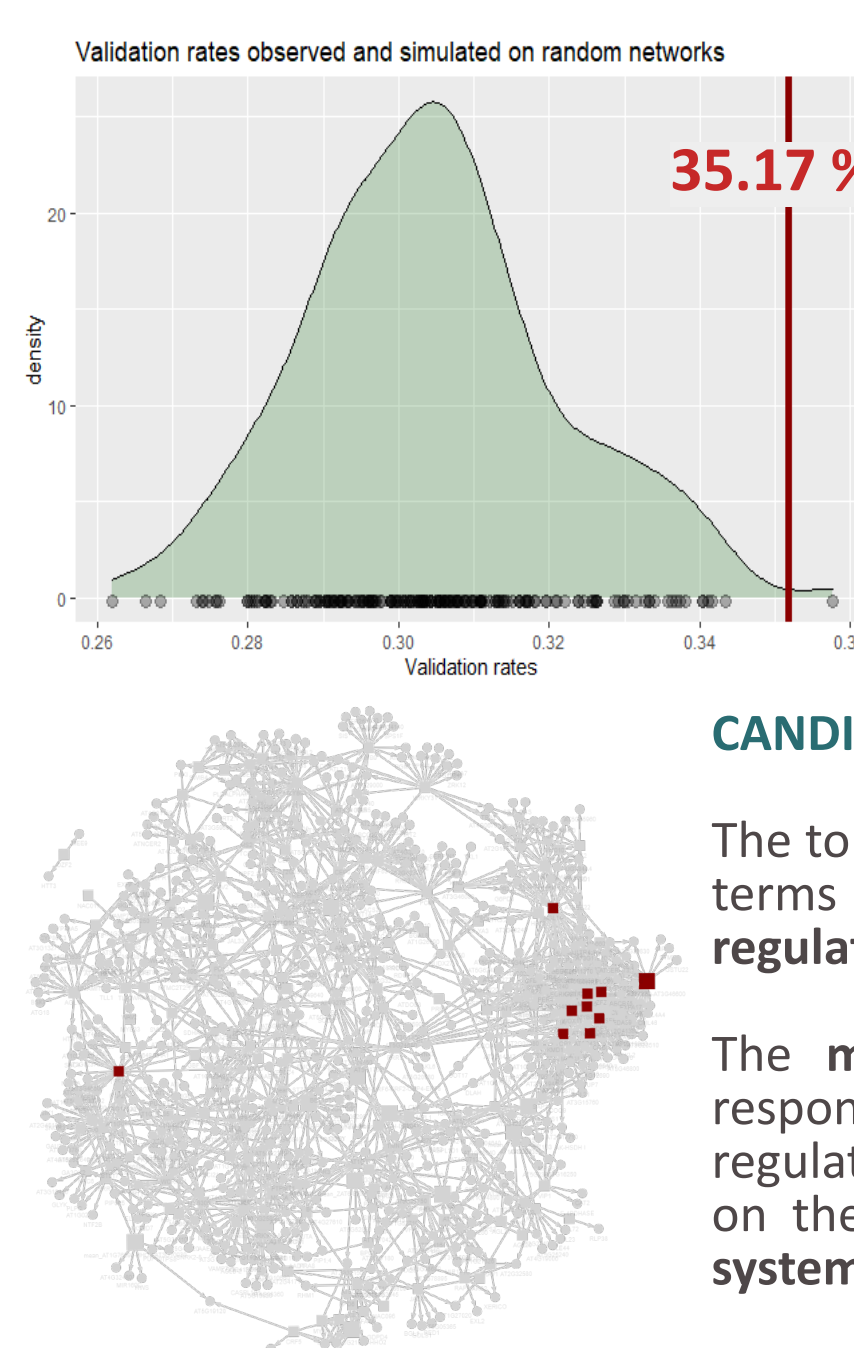
Machine learning techniques described in section 04 were used on the CN vs N genes, to infer transcriptional dependencies between regulators and targets based on their expression profiles across perturbations of nitrate and CO₂

NETWORK RECONSTRUCTION PIPELINE



NETWORK ENRICHMENT AND VALIDATION

- We could confirm predicted associations between **nitrate transport** genes and some of their regulators (BT1-BT2-HRS1 grouped together, NRT2.1-NAR2.1 co-regulated...)
- To validate the inferred connections, we made use of **connectTF**^[7], a recent database containing regulatory interactions in Arabidopsis thaliana obtained from in vitro (DAP-Seq) and in vivo (CHIP-Seq) binding experiments, as well as in planta regulation experiments (TARGET).



35.17 % of the predicted regulatory pairs with validation data available were true positives. We simulated shuffled networks with identical nodes and observed that our inference technique has a significant added value in terms of **prediction precision** as compared to randomness.

Validation rates were computed for 200 random networks, and compared in an empirical statistical test to the observed validation rate, $p = 0.005$

CANDIDATE REGULATORS IDENTIFICATION

The top 25 % of most connected regulators were found enriched in GO terms linked to gene expression regulation, as well as **negative regulation of nitrate metabolism**, and **root system plasticity**.

The **most connected genes** are potential **key regulators** of the response, and represent genes of interest. The ten most connected regulators of the CO₂ and nitrate perturbations network - **red** nodes on the left network - were selected, and are currently under **root system and shoots mineral status phenotyping**.

BEHAVIOR OF GENES DIFFERENTIALLY EXPRESSED BY ELEVATED CO2 UNDER LOW NITRATE

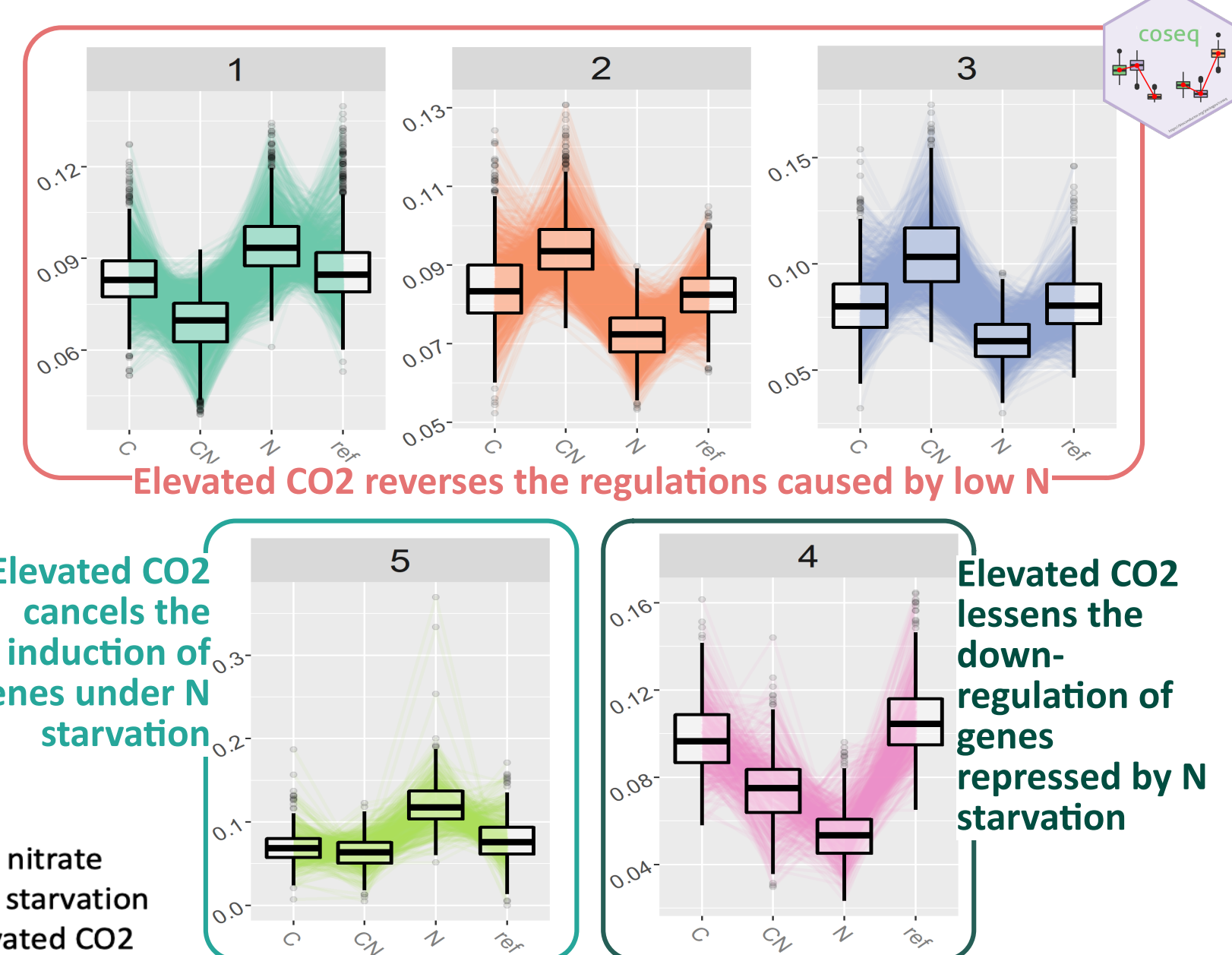
Poisson Mixture Models were adjusted via the Coseq R package to group genes with similar expression profiles across perturbations of nitrate and CO₂

ELEVATED CO2 AND LOW NITRATE

By comparing the **conditions CN and N to ref** in the above gene cluster profiles, we can understand how elevated CO₂ affects plants that are under low nitrate conditions, as compared to reference levels. In those clusters, **elevated CO₂ always reprograms gene expression in an opposite way** as observed in the regulations at stake in simple nitrate starvation.

ELEVATED CO2 AND IRON STARVATION

Very similar **disruptions of the response to iron starvation** were observed with this method, by running gene clustering on the conditions ref, C, F and CF.



Expression profiles obtained by clustering the CN vs N genes on the conditions ref, C, N and CN

05. Conclusions

- There are clear **interactions between high atmospheric CO₂ and nutritional starvations**, that can not be explained by the addition of individual perturbations

- The deleterious effect on mineral composition and the reprogramming of gene expression by elevated CO₂ are almost **canceled when plants grow in nutrient-rich media**. This supports previous findings in the literature on CO₂ acclimation, lifted by nitrate rich nutrition^[8].

- Gene-level analyses as well as genome-wide studies such as clustering or network inference showed that **disrupted root architecture and nutritional strategies in roots** could explain the degradation of nutritional value in plants under climate change

- Experimentally testing candidate genes for their implication in deleterious phenotypic response under high CO₂ could be a **promising lead** for crops improvement in the future.

PERSPECTIVES

- Studying shoot transcriptomes would allow to understand CO₂ effect on photosynthesis efficiency, as well as suspected shoot to root signals

- Time series or pseudo longitudinal transcriptomes under CO₂ exposure could reveal more causal gene interactions and the dynamic of regulations at stake in this response

- GWAS studies could reveal genomic variants influencing plant mineral status under elevated CO₂ and suggest new candidate genes

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